



MILESTONE 3 · GROUP 8 · IITM DS & AI LAB

AI-Powered Wildfire Early Detection

Model Architecture & End-to-End Pipeline

Problem Statement & M3 Goal

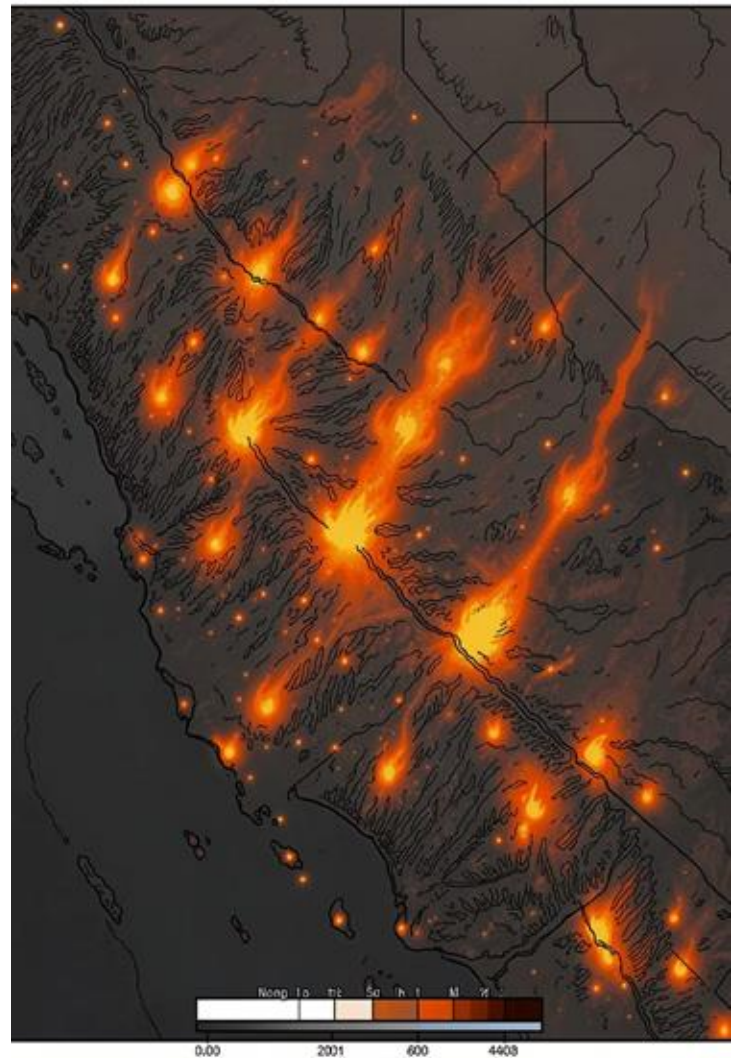
Predict next-day wildfire risk over California using multi-source EO + weather. Primary unit: ERA5 0.25° cell × day (FIRMS conf $\geq 30 \rightarrow$ fire label).

Core Task

Predict next-day wildfire risk over California using multi-source EO + weather

M3 Deliverable

Architecture + baseline ladder + training + released checkpoints + maps



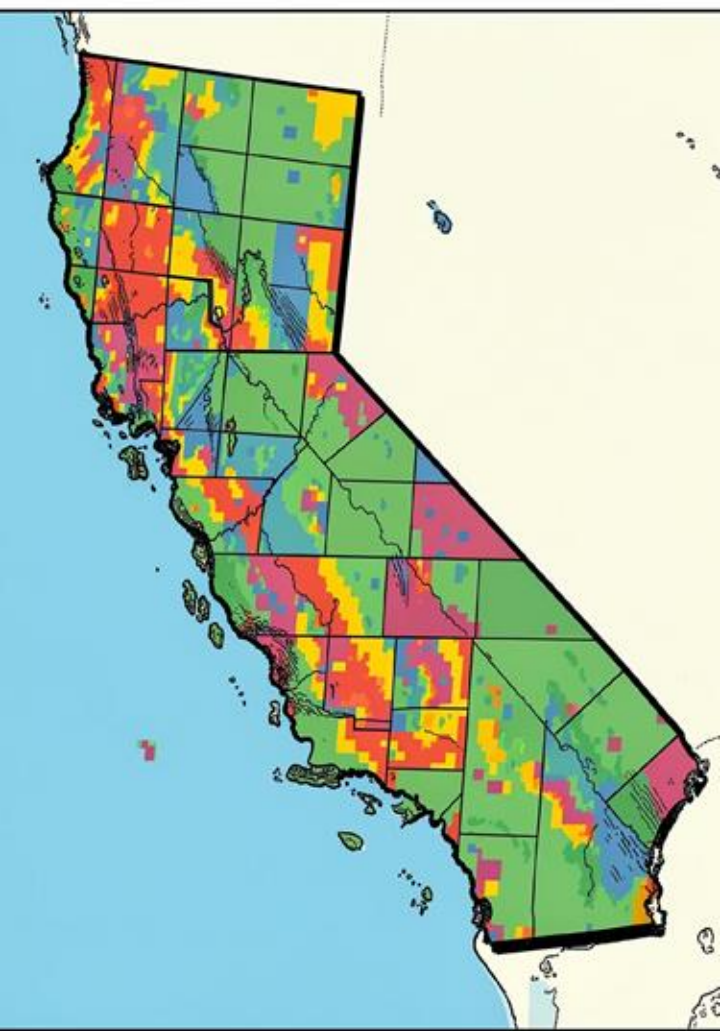
Two Sibling Stacks

	Primary fusion	Experiments
Task	Cell-day next-day classification	Patch per-pixel segmentation
Grid	ERA5 0.25°	FIRMS ~1 km, 64×64
Split	2022–23 / 2024 / 2025	2025 temporal 70/15/15
Headline	Test PR-AUC 0.531	Test F1 0.110 (ConvLSTM)

Same sensors family; different task — numbers are not 1:1 comparable.

Architecture Ladder (Primary)

- mvp_era5_dem → LightGBM (ERA5 + DEM)
- cnn_s2_mvp → S2 CNN + weather MLP
- cnn_lstm_fusion → S2 CNN + 7-day LSTM ± S5P
- multimodal_fusion → five-branch hybrid (primary)



MultimodalFusion Architecture

- S2 CNN ($6 \times 64 \times 64 \rightarrow 128\text{-d}$) + S5P CNN ($2 \times 64 \times 64 \rightarrow 64\text{-d}$)
- LSTM on ERA5+DEM $7 \times 27 \rightarrow 64\text{-d}$
- S2 / S5P numerical MLPs $\rightarrow 64\text{-d} / 32\text{-d}$
- Concat $\rightarrow$ dense head $\rightarrow$ next-day fire logit
- Train: BCE + pos_weight · early-stop on val PR-AUC · isotonic calibration

Data & Pipeline (Primary)

- GCS: ERA5, FIRMS, S2/S5P mosaics + numerical CSVs, DEM cell table
- Build: tabular → patches → sequences → numerical joins → train → maps
- Temporal split (forecast-safe); fire season May–Nov
- outputs/ local / gitignored · small artifacts/ in git

Released Results (Primary)

CNN + LSTM + S5P

Split	ROC-AUC	PR-AUC
Test calibrated	0.805	0.489

MultimodalFusion (all branches)

Split	ROC-AUC	PR-AUC
Val raw (best)	0.832	0.569
Test calibrated	0.831	0.531

Multimodal beats CNN+LSTM on test PR-AUC (0.531 vs 0.489)

Complementary Experiments (Patch Stack)

- Location: Milestone 3/Experiments/ · Apple Silicon MPS
- 30 fused channels · fire-centered 64×64 · 1035 patches (2025 Jan–Nov)
- Models: HistGB · ConvLSTM+U-Net · U-Net last-day · BCE+Dice / Focal
- Offline pack A–D + confusion matrices

Default Config

hidden=32 · lr=3e-4 · batch=4 · loss=bce_dice ·
device=mps

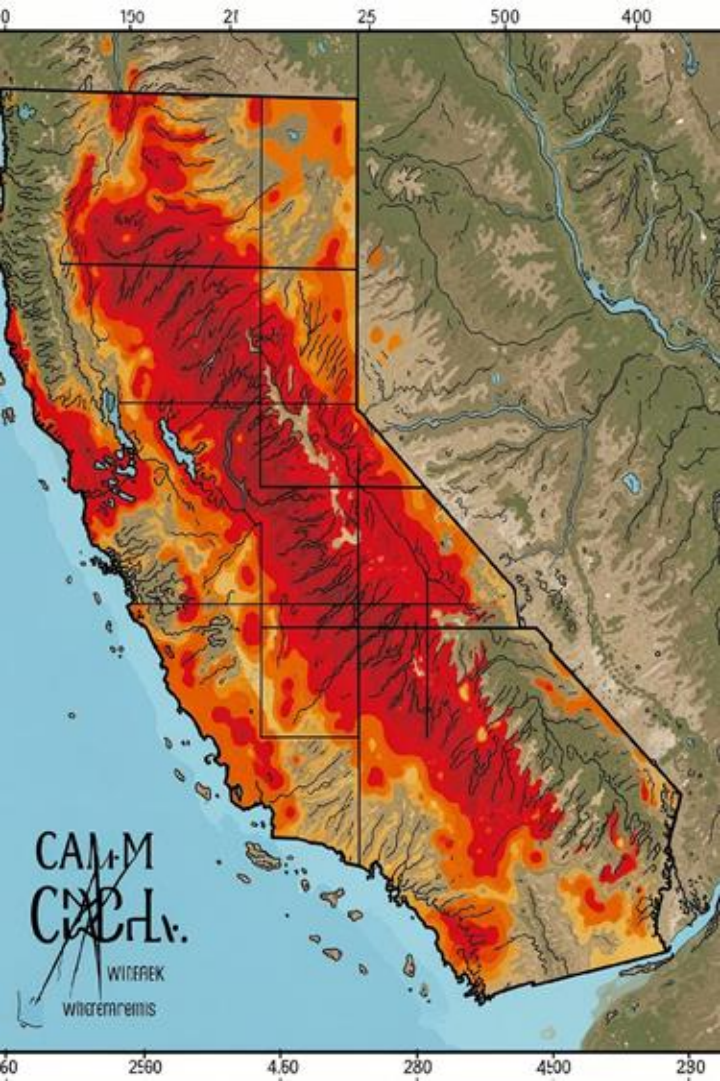
Headline Result

DL test F1 = 0.110 vs baseline 0.029 — a 3.8× improvement on the same temporal split. Val AUC-PR: 0.359 (DL) vs 0.070 (baseline).

Experiments Results (Test)

ID	Model	F1	AUC-PR
A	HistGB	0.029	0.009
B	ConvLSTM + BCE+Dice	0.110	0.046
C	ConvLSTM + Focal	0.025	0.065
D	U-Net last-day	0.086	0.040

B best F1; C best ranking but low recall at 0.5. Late-season sparse fires — accuracy not a headline metric.



Key Decisions & Closing

Key Decisions

Temporal splits; cell×day for ops-style alerts; calibrated confidence %. Progressive fusion ladder; git small artifacts only. Experiments validate dense FIRMS-grid segmentation on a 2025 candidate.

Limitations

Seasonality / threshold transfer; mosaic timing; not operational alerts yet.

Milestone 3 — Complete

Architecture + E2E pipeline + released multimodal weights. Headline (cell-day): test calibrated PR-AUC 0.531 (ROC 0.831). Complementary: patch ConvLSTM test F1 0.110 vs baseline 0.029.

Team: Ripunjay · Lakshay · Roushan · Lakshmi Sruthi · Aditya

Member	Roll Number	Signature
Ripunjay Kumar	21F3002511	✓
Lakshay Garg	21F3001076	✓
Roushan Kumar Singh	23F1002240	✓
Lakshmi Sruthi K	21F1005626	✓
R Aditya	21F1004839	✓